

The Price of Reanalysis: Varying Ambiguous-Region Length in NP/Z Garden Paths to Dissociate Structural from Linear Prediction

Oscar Dalton
Swarthmore College

odalton1 at swarthmore dot edu

Abstract

Human sentence processing is organized around hierarchical structure rather than linear word order. Transformer language models carry no built-in structural bias and learn from flat token sequences, yet a well-known line of work shows that their surprisal predicts garden-path difficulty while badly underestimating its size (van Schijndel & Linzen, 2021; Huang et al., 2024). One reading of that gap is that the models lack structure. Another is that they have it and leave it idle until something forces its use. Separating these requires a design in which hierarchical relationship and linear distance vary independently, and I report a behavioral experiment built on one. In an NP/Z paradigm adapted from the SAP Benchmark (Huang et al., 2024), the critical word and the structural relationship stay fixed while the temporarily ambiguous region is lengthened. GPT-2-large and Pythia-1.4B both show a large garden-path effect (5.3 to 6.5 bits) at every distance, and the effect stays flat as the disambiguator moves further from the material it revises: confidence intervals bound any growth well below the smallest change I treat as meaningful. The presence of the effect rules out prediction from a context window too short to reach the disambiguating comma. Its failure to grow is hard to square with recency-weighted linear prediction, and it is what a structure-governed account expects. Humans behave differently: their reanalysis difficulty grows with the length of the ambiguous region (Ferreira & Henderson, 1991; Tabor & Hutchins, 2004), so a model without that length penalty will fall short of human difficulty by a margin that widens with distance, which gives the underprediction result a length dimension it did not previously have. The experiment establishes the behavioral pattern an encoded-but-unused account requires, though it cannot by itself distinguish that account from full structural governance; that distinction needs the probing and intervention work sketched at the end.

1 Introduction and Motivation

Natural-language dependencies are defined over constituency, not surface position: agreement, binding, and filler-gap relations track hierarchical structure no matter how many words intervene. Transformer language models acquire these dependencies with striking fidelity despite carrying no built-in

notion of a tree; they read over flat token sequences. This poses a question that a decade of probing has circled without settling: do these models recover the hierarchical structure that governs human language, or do they approximate it with linear statistics good enough to pass most benchmarks?

One line of evidence makes the question concrete. Using surprisal as the linking function between a model’s next-word probabilities and human reading times (Hale, 2001; Levy, 2008; Futrell et al., 2019), work from the Linzen lab established a robust and uncomfortable result: single-stage language-model surprisal correctly predicts that garden-path sentences cause processing difficulty, but badly underestimates how much (van Schijndel & Linzen, 2021), and at scale, surprisal fails to predict even the relative difficulty of different ambiguity types (Huang et al., 2024). The standard reading is that something structural is missing from the model. Two recent results complicate that conclusion. Ahuja et al. (2025) show that hierarchical and linear subnetworks coexist within a single trained transformer, and Michaelov & Arnett (2025) show that model behavior on agreement passes through frequency- and locality-driven phases before settling into generalized, structure-sensitive rules. Together they point toward a different diagnosis: the structure may be there, sitting idle whenever a linear shortcut gets the right answer.

Garden-path reanalysis is the cleanest available probe of that diagnosis. A temporary ambiguity invites a locally driven reader to commit to one parse; a disambiguating word later forces a structure-driven revision, and the competing processing accounts diverge at that word. Reanalysis is the moment structure and linear order come apart, so it is the natural place to ask whether a model that can represent the correct structure consults it before being forced to. I study one construction in depth, the NP/Z (noun-phrase/zero) ambiguity, the most controllable of the family; an NP/S replication and a converging agreement-attraction paradigm are planned to show the result is not tied to a single structure.

Prior surprisal work, for all its precision, does not orthogonalize linear distance from structural relationship within its items, so a behavioral effect cannot by itself separate a linear from a hierarchical explanation. The contribution of this paper rests on a single methodological move that repairs this. I introduce a stimulus design that crosses hierarchical relation-

ship with linear distance: the critical word and the structural relationship are held fixed while the temporarily ambiguous region is lengthened with plausibility-screened modifiers. No individual item confounds the two factors, so a structural and a linear account of the garden-path effect must part ways in their predictions, and the design transfers unchanged to other constructions, models, and languages. Deriving those predictions takes more care than the intuitive gloss suggests, because a transformer is not a human memory system. Section 3 does the derivation explicitly, and in the course of it the linear account splits in two; the presence of the garden-path effect refutes one variant, and its invariance refutes the other.

The empirical result, reported in Section 6, is that on two open-weights models from different families, GPT-2-large and Pythia-1.4B, the garden-path effect at the disambiguating word is large at every distance and does not grow as the disambiguator moves further from the material it forces a revision of. A structure-governed account predicts this. Human comprehenders do the opposite: their reanalysis difficulty grows with the length of the ambiguous region (Ferreira & Henderson, 1991; Tabor & Hutchins, 2004), the digging-in effect, so the models’ lack of a distance penalty marks a departure from human processing, not a convergence with it. I argue in Section 7 that the departure makes the underprediction puzzle more specific. A comprehender whose commitment to the wrong parse deepens with time will show a reanalysis cost that grows with distance, and a model with no such dynamic will fall short of that cost by a margin that grows with distance too. That is a refinement of the underprediction result with empirical bite of its own.

The behavioral experiment cannot, by design, distinguish a model that consults structure throughout from one in which structure sits in the hidden states and gets consulted only when a disambiguator forces the issue; both predict the invariant behavior observed. Telling them apart requires reading structure out of internal states with probes, establishing what the output actually depends on through causal intervention, and charting when, over training, decodability and use come apart. Section 9 sketches that program briefly; the claims of the present paper are confined to what the behavioral data show.

The remainder of the paper proceeds as follows. Section 2 situates the work within the surprisal-underprediction and hierarchical-bias literatures; Section 3 derives the competing predictions; Section 4 develops the dissociation design; Section 5 details the models, the surprisal computation, and the analysis; Section 6 reports the result; Section 7 takes up the human comparison; Section 8 addresses threats to validity; and Section 9 states the significance and the program that follows.

2 Background and Related Work

2.1 Surprisal theory and language models as psycholinguistic subjects

The empirical claims of this paper are read off a single linking function. Expectation-based theories of comprehension hold that the difficulty of processing a word is proportional to its surprisal, the negative log probability that a comprehender’s language model assigns to that word in context (Hale, 2001; Levy, 2008). Surprisal theory makes processing difficulty a function of prediction alone, which is what licenses substituting a language model for the human comprehender: the model’s word-by-word surprisal becomes a candidate predictor of word-by-word reading time. Futrell et al. (2019) make this substitution explicit, treating neural language models as psycholinguistic subjects and probing their expectations about syntactic state with the same targeted sentences a psycholinguist would administer to people. I adopt surprisal as the behavioral linking function throughout. The strength of this commitment is also its exposure: if a model’s surprisal diverges from human difficulty in a structured way, that divergence is itself a finding about what the model is and is not computing, and it is the lever the garden-path literature pulls.

2.2 Hierarchy versus linear order in neural models

Whether neural models represent the hierarchical structure of language or approximate it with linear order has been contested since before transformers. Targeted agreement tasks were the first probe: Linzen et al. (2016) test whether a model tracks subject-verb number across intervening material, treating success on structure-sensitive dependencies as evidence of structural representation. The picture that emerged is that linear order is the default. McCoy et al. (2020) show that recurrent networks trained on data consistent with both a hierarchical and a linear rule generalize linearly unless the architecture or training signal pushes them otherwise, and Petty & Frank (2021) find the same linear preference in transformers on analogous transformation tasks. Two results qualify this default in ways that matter here. Murty et al. (2023) find that hierarchical generalization can emerge only after long training, well past the point where in-distribution accuracy saturates, a late and abrupt transition they term structural grokking. And most directly, Ahuja et al. (2025) report that hierarchical and linear subnetworks coexist within a single trained transformer rather than one replacing the other. The coexistence matters conceptually: if both solutions live in one network, asking whether structure is represented becomes the wrong question, and the live question becomes which subnetwork governs the output, and when.

2.3 Garden paths, digging in, and the underprediction puzzle

This is the line this paper extends. The garden-path constructions, and the minimal-attachment account of why they are

hard, come from the classical sentence-processing literature (Frazier & Rayner, 1982): a reader pursues the locally preferred parse, reaches a disambiguating word inconsistent with it, and incurs a measurable cost to reanalyze. Two properties of the human effect matter for what follows. The first is that it is not distance-invariant: lengthening the ambiguous region makes recovery harder, an effect established by Ferreira & Henderson (1991) and developed by Tabor & Hutchins (2004) as digging in, on the interpretation that the comprehender’s commitment to the misparse strengthens as evidence consistent with it accumulates. Model-human comparisons of garden-path behavior therefore involve length as well as magnitude. The second property is the underprediction result. van Schijndel & Linzen (2021) show that while language-model surprisal predicts that garden-path sentences are harder, the slowdown it predicts is far smaller than the slowdown humans show, by a margin large enough that closing it would demand implausible surprisal-to-reading-time scaling. Arehalli et al. (2022) ask whether structurally informed surprisal repairs the shortfall and find that it narrows the gap without closing it. Huang et al. (2024), drawing on self-paced reading from two thousand participants across a range of constructions, push the result further: surprisal underestimates the magnitude of disambiguation difficulty and also gets its relative ordering across construction types wrong, which is difficult to reconcile with any account on which surprisal is the whole story. The standard reading of this body of work is that something structural is missing from the model. Michaelov & Arnett (2025) complicate that reading from the training-dynamics side, showing that model behavior on agreement passes through frequency-driven and locality-driven phases before it settles into generalized, structure-sensitive rules. A model partway along that trajectory would encode structure it does not yet consistently deploy, the encoded-but-not-yet-used situation this project sets out to test.

2.4 Separating encoded from used structure

Establishing that structure is encoded requires reading it out of the model’s internal states, and the methods for doing so carry a caveat that shapes the larger program this paper opens. Structural probes recover syntactic structure from contextual representations (Hewitt & Manning, 2019), but Hewitt & Liang (2019) show that a sufficiently expressive probe can recover structure the model never uses for its predictions, so a probe result alone cannot distinguish encoded-and-used from encoded-but-inert, which is the distinction at issue. Causal interventions on internal activations (Meng et al., 2022) test whether a representation makes a difference to the output rather than merely being decodable from it, and suites with released training checkpoints, most cleanly Pythia (Biderman et al., 2023), make it possible to watch decodability and use emerge separately over training. These methods define the second phase of this project and are not applied here; the present paper establishes the behavioral pattern that makes

them worth applying.

2.5 The gap

Each of these literatures supplies a piece, and none combines them. The agreement and transformation work establishes that linear order is the default and that hierarchical structure may coexist with it, but it reads structure off behavior or static probes rather than at the moment of reanalysis. The surprisal-underprediction work localizes a behavioral failure but cannot say whether the missing structure is absent or merely unconsulted, and it has not manipulated the length of the ambiguous region within items, so the digging-in dimension of the human effect has no model-side counterpart. No prior study uses garden-path reanalysis with linear distance orthogonalized from hierarchical relationship. The hypotheses in Section 3 are organized around closing that gap.

3 Hypotheses and Predictions

Deriving what a linear account predicts requires saying what linear prediction means in a transformer, and the answer is not unique. In a human comprehender, the prediction that reanalysis cost grows with distance falls out of memory architecture: representations decay and retrieval suffers interference, so revising material further back is harder. A transformer has no such decay; attention is content-addressed, and a full-attention model can retrieve any earlier token equally well at any of the distances used here. The linear hypothesis therefore has to be stated as a claim about the effective context the model’s next-word distribution depends on, and it splits into two variants with different empirical commitments.

3.1 H1a, strictly local prediction

On the strictly local variant, the output distribution at each position depends only on a short window of preceding tokens, in the manner of an n -gram model. This variant is refuted not by the distance manipulation but by the garden-path effect itself. In the materials used here (Section 4), the ambiguous and control conditions differ only in a comma that falls outside any short window around the critical word: at the critical word, the immediately preceding tokens (“...the file deserved”) are identical across conditions. A predictor whose effective context does not reach the comma therefore assigns the same probability to the critical word in both conditions and predicts no garden-path effect at all. A large effect is direct evidence that the model’s prediction at the critical word depends on material beyond the local window, including the presence or absence of the disambiguating comma several words back.

3.2 H1b, recency-weighted linear prediction

On the recency-weighted variant, the model’s prediction depends on the full context but weights it by linear proximity, whether through learned positional biases, induction-style copying of recent material, or any mechanism that privileges

nearby tokens over distant ones. This is the variant for which distance genuinely matters. As the ambiguous region lengthens, the comma (or its absence) and the subordinate verb recede from the critical word while the locally supportive material (the noun phrase read as a plausible object) sits adjacent to it, so the influence of the disambiguating evidence is diluted relative to the influence of the misleading local evidence. H1b therefore predicts that the garden-path effect changes systematically with distance: the surprisal gap between ambiguous and control conditions grows as the revision-forcing relationship spans more tokens, because the recency-weighted reader is increasingly captured by the local parse. In the dissociation design, where the critical word and the structural relationship are held fixed while the ambiguous region is lengthened, H1b predicts a positive ambiguity by distance interaction.

3.3 H2, the structure-governed account

On the structure-governed account, the model resolves the ambiguity over the structural relationship between the relevant words, and the cost of reanalysis depends on that relationship rather than on how many words intervene. The prediction for the garden-path constructions is that the effect is invariant across linear distance once structure is held fixed: lengthening the ambiguous region with structurally inert material leaves the reanalysis cost at the critical word essentially unchanged. The earlier literature invites a conflation here that is worth heading off. Invariance is what a structural account predicts for a transformer, and it is not what humans do: human reanalysis difficulty grows with the length of the ambiguous region (Ferreira & Henderson, 1991; Tabor & Hutchins, 2004), so a model matching H2 diverges from human comprehenders on the very dimension this design manipulates. H2 is the direct contrast to H1b, and because the design makes the two factors vary independently within items, a single behavioral experiment separates them.

3.4 H3, the hybrid encoded-but-not-used account

H3 is the account this project predicts. It does not sit between H1 and H2 on a continuum; it concerns the relationship between what the model represents and what drives its output. On this account, the hierarchical structure is recoverable from the model’s hidden states throughout the ambiguous region, while the output distribution is dominated by linear, locally driven prediction until the disambiguating word forces a revision: structure is encoded but not used. The motivation is the coexistence of hierarchical and linear subnetworks within single trained transformers (Ahuja et al., 2025) and the locality-driven phases that precede structure-sensitive behavior over training (Michaelov & Arnett, 2025). Behaviorally, H3 and H2 predict the same distance-invariant outcome, since on both accounts the revision, once triggered, is computed over the structural relationship; what separates them is internal, and the two accounts come apart only under probing and causal intervention. The behavioral experiment reported here

can therefore refute H1a and H1b and establish the behavior common to H2 and H3, but adjudicating between H2 and H3 is the work of the second phase, along with the developmental prediction that follows from H3: if structure is encoded before it is used, training checkpoints should show a period in which a structural probe succeeds while the output-level structural correction is weak, a lag that narrows with training and scale.

4 Materials: The Dissociation Design

The design principle is to cross hierarchical relationship with linear distance so that no single item confounds them. The garden-path effect is defined as surprisal in the ambiguous condition minus surprisal in the matched control at the critical word; subtracting the control removes the length baseline, so any residual distance effect on the difference score is attributable to the interaction of ambiguity with distance rather than to sentence length itself.

The primary construction is the NP/Z (noun-phrase/zero) ambiguity. In the ambiguous condition, a subordinate-clause verb can be parsed transitively, taking the following noun phrase as its object (“Because the suspect changed the file . . .”), until a verb arrives that requires that noun phrase as its subject (“... deserved further investigation”), forcing reanalysis. In the control condition a comma after the subordinate verb (“Because the suspect changed, the file . . .”) blocks the transitive parse and no reanalysis is required. The critical word is the disambiguating verb, identical across conditions, and the material immediately preceding it is likewise identical across conditions; this identity is the source of H1a’s zero-effect prediction.

Items were adapted, not authored: the base set is the NP/Z subset of the SAP Benchmark (Huang et al., 2024), comprising twenty-four lexicalizations. The distance manipulation lengthens the temporarily ambiguous noun phrase with zero, one, or two prenominal adjectives (“the file,” “the confidential file,” “the confidential legal file”), holding the critical word, the disambiguating relationship, and the postcritical spillover material fixed. Prenominal modification was chosen over postnominal alternatives (prepositional phrases, relative clauses) because postnominal modifiers would introduce a second attachment ambiguity and thereby contaminate the manipulation; the cost of this choice is that the manipulation spans a narrow range, at most two added tokens, a limitation taken up in Section 8. Added modifiers were screened for plausibility drift with a language-model scorer; the screening model was GPT-2-large, which subsequently also served as one of the two analysis models, a circularity discussed in Section 8. The full design is twenty-four items $\times$ two conditions (ambiguous, control) $\times$ three distances (d1, d2, d3), yielding 144 sentences. The item set and the analysis plan, including the confirmatory test, were fixed before any model was run.

5 Methods

5.1 Models

The behavioral experiment uses two open-weights models: GPT-2-large (774M parameters; Radford et al., 2019) and Pythia-1.4B (Biderman et al., 2023). The two differ in architecture family, training data, and tokenizer, so agreement between them guards against model-specific artifacts. Llama-3 8B, the intended modern reference point and the target of the planned mechanistic work, is deferred pending compute and access constraints; the dissociation design and the pipeline transfer to it unchanged. All weights are accessed through Hugging Face with pinned revisions; probabilities are read directly and no sampling is performed.

5.2 Surprisal computation

The surprisal of word w_i in context is $-\log_2 p(w_i | w_{<i})$, in bits. Per-token surprisal is computed with the minicons library (Misra, 2022). Because both models use subword tokenization, critical words are aligned to token spans explicitly: the critical word’s character span is located in the sentence, mapped to token indices through the tokenizer’s offset mapping, and surprisal is summed over the tokens in the span when a word is split. The alignment was validated by hand on five items per model, and the per-row token counts recorded by the pipeline confirm that every critical word tokenized to a single subword in both tokenizers, with no alignment failures (144 of 144 sentences scored per model).

5.3 Behavioral analysis

The confirmatory test, fixed in advance, is the ambiguity $\times$ distance interaction on critical-word surprisal. Because the two models score the same 144 sentences, their observations are not independent, and pooling them without a model term would treat each sentence as two independent data points. The primary analysis is therefore conducted per model: for each model separately, a linear mixed-effects model with fixed effects for ambiguity (ambiguous versus control), linear distance (coded numerically and centered at its mean, so that the ambiguity coefficient is evaluated within the observed range), and their interaction, a random intercept for item, and a by-item random slope for ambiguity where the fit converges (statsmodels MixedLM, REML). A pooled model with a fixed effect of model family and random intercepts for item is reported as a secondary analysis. Because the theoretically important outcome is a null interaction, significance testing is supplemented with estimation. I report the two-sided 95% confidence interval on the interaction, expressed throughout as the change in the garden-path effect (ambiguous minus control) per distance step, and assess it against a smallest effect size of interest of 0.3 bits per step, chosen as ten percent of the roughly six-bit ambiguity main effect over the full two-step range. Because the claim under test is one-sided, that the effect does not grow, the relevant quantity is the upper bound of that interval: the

Model	Condition	d1	d2	d3
GPT-2-large	ambiguous	17.38	16.77	16.49
	control	10.87	10.85	10.58
	GP effect	6.51	5.93	5.91
Pythia-1.4B	ambiguous	17.24	16.71	16.28
	control	11.37	11.46	10.79
	GP effect	5.87	5.25	5.49

Table 1: Mean critical-word surprisal (bits) by model, condition, and distance, with the garden-path (GP) effect defined as ambiguous minus control. $n = 24$ items per cell. GP effects are computed from unrounded cell means and rounded independently, so a displayed effect may differ from the difference of the displayed means by 0.01.

upper limit of a two-sided 95% interval is a one-sided bound at $\alpha = .025$, which is conservative relative to the $\alpha = .05$ bound that defines the standard two one-sided tests procedure for equivalence (Lakens, 2017). Under H1b the garden-path effect grows with distance; under H2 and H3 it does not, with the upper confidence bound excluding growth at or beyond the smallest effect size of interest.

6 Results

6.1 The garden-path effect is large in both models

Table 1 reports mean critical-word surprisal by condition and distance for each model. The garden-path effect, ambiguous minus control at the critical word, is large at every distance in both models, ranging from 5.25 to 6.51 bits. In the per-model mixed models, with distance centered so that the coefficient is evaluated at the middle of the tested range rather than extrapolated beyond it, the main effect of ambiguity, reported in the ambiguous-minus-control direction used throughout, is 6.11 bits for GPT-2-large ($SE = 0.55$, $p < .001$) and 5.54 bits for Pythia-1.4B ($SE = 0.55$, $p < .001$); the by-item random slope for ambiguity converged in both fits. This effect alone carries inferential weight: because the ambiguous and control conditions are identical in the tokens immediately preceding the critical word, a predictor whose effective context excluded the disambiguating comma would show no effect at all (H1a, Section 3). A five to six bit effect, a factor of roughly forty to ninety in probability, refutes strictly local prediction outright: both models’ expectations at the critical word are conditioned on material well beyond the local window.

6.2 The effect does not scale with linear distance

The confirmatory test is the ambiguity $\times$ distance interaction, expressed as the change in the garden-path effect per distance step. It is not significant in either model (GPT-2-large: -0.30 bits per step, 95% CI $[-0.64, +0.04]$, $p = .083$; Pythia-1.4B: -0.19 bits per step, 95% CI $[-0.54, +0.15]$, $p = .279$), nor in the pooled secondary analysis (-0.25 bits per step, 95% CI $[-0.73, +0.24]$, $p = .315$): the garden-path effect does not grow across the distance manipulation (Figure 1). Treated as

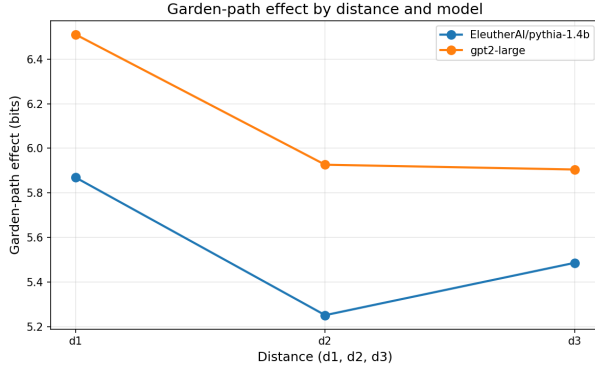


Figure 1: Garden-path effect (ambiguous minus control, in bits) at the critical word as a function of linear distance, per model. The effect is large at every distance and does not grow with distance, contrary to the growth predicted by recency-weighted linear prediction (H1b).

an estimation problem rather than a significance test, the confidence intervals are asymmetric in a way that matters. Growth is what H1b predicts, and growth is tightly bounded: at most $+0.04$ bits per step in GPT-2-large and $+0.15$ in Pythia-1.4B, both well under the smallest effect size of interest of 0.3 bits per step. These upper limits are one-sided bounds at $\alpha = .025$, stricter than the $\alpha = .05$ bound conventional in equivalence testing (Lakens, 2017), so the exclusion of the smallest effect size of interest holds a fortiori under the standard procedure. The data therefore exclude H1b’s prediction at any magnitude worth defending, in both models. Shrinkage is less tightly bounded, with effects as large as -0.64 bits per step compatible with the GPT-2-large interval, and the nonsignificant trend in that model runs toward shrinkage; the invariance claim is accordingly one-sided. The data rule out growth, the direction that carries the theoretical weight; they do not rule out every form of distance modulation. A main effect of distance is present in both models (-0.29 and -0.39 bits per step for GPT-2-large and Pythia-1.4B, both $p < .001$), such that surprisal at the critical word declines modestly with added prenominal material in ambiguous and control sentences alike, a pattern consistent with the added adjectives making the head noun and its continuation more predictable. The interaction term tests whether this decline differs between conditions, and it does not.

The result bears on the hypotheses as follows. H1a is refuted by the presence of the effect, as argued above. H1b predicts that the garden-path effect grows as the disambiguator moves further from the material it forces a revision of; the observed trend runs, if anything, in the opposite direction, marginally so in GPT-2-large, and growth is bounded below the smallest effect size of interest in both models. A flat effect across distance is what H2 and H3 both predict. Two cautions attach to this reading. The critical result is a null interaction over a distance range of at most two added tokens, and with

twenty-four items the design has limited power against a small scaling effect; the equivalence analysis above states how small an effect the data can and cannot rule out, and Section 8 returns to the narrowness of the manipulation. And the behavioral data cannot distinguish H2 from H3, which make identical predictions at the output; the encoded-versus-used question is answerable only with probing and intervention, so the present result is consistent with, but not confirmatory of, H3.

6.3 Robustness

The result replicates within the item set. Splitting the twenty-four items into random halves and refitting the per-model specification on each, the ambiguity effect is large and significant in every cell (6.86 and 5.37 bits for GPT-2-large, 5.95 and 5.13 for Pythia-1.4B, ambiguous minus control at mean distance, all $p < .001$). The interaction is significantly positive in no half; the only significant interaction anywhere is a shrinking effect in one Pythia-1.4B half (-0.52 bits per step, $p = .041$), the direction opposite the growth H1b predicts, and the remaining halves are null ($p = .18$ to $.55$). With twelve items the halves are too small to bound growth tightly on their own, so the equivalence claim rests on the full-sample intervals in Section 6.2; the halves establish that the presence, size, and direction of the effect are not carried by any particular subset of items. The direction of the effect, ambiguous exceeding control, holds in both models individually, and the two models yield the same qualitative pattern despite different architectures, training corpora, and tokenizers. In the present materials tokenization is not a plausible artifact source, since every critical word aligned to a single subword token in both tokenizers.

7 Discussion: Divergence from the Human Profile

The absence of a distance penalty is the outcome the design was built to detect, and it is the opposite of what human comprehenders do. Human recovery from a garden path is harder when the ambiguous region is longer (Ferreira & Henderson, 1991), and Tabor & Hutchins (2004) interpret this digging-in effect as the comprehender’s commitment to the misparse strengthening as locally consistent evidence accumulates. The models show no trace of this dynamic. Their revision cost at the disambiguator does not grow with how long the misleading parse was entertained, and the only significant or near-significant trends in the data, a marginal tendency in GPT-2-large and one significant split-half interaction in Pythia-1.4B, run toward a smaller cost at longer distances, the reverse of digging in.

The divergence gives the underprediction puzzle a more specific shape. If human garden-path difficulty has a component that grows with the length of the ambiguous region and model surprisal does not, then surprisal will underpredict human difficulty by a margin that itself grows with length, and

the underprediction established at the aggregate level by van Schijndel & Linzen (2021) and Huang et al. (2024) should be decomposable into a structural component the models capture and a commitment-or-memory component they lack entirely. This is a testable claim: the SAP Benchmark releases the human self-paced reading data for the items adapted here, and regressing the human-model gap on ambiguous region length is a direct test. That analysis is beyond the scope of the present paper but is the natural next behavioral step, and it changes the question this line of work should be asking: the issue stops being how much difficulty surprisal misses and becomes which kind.

8 Threats to Validity and Limitations

The most important limitation is the scale of the evidence. The behavioral result rests on one construction (NP/Z), twenty-four adapted items, and two models below two billion parameters; the NP/S replication, the agreement-attraction converging paradigm, and the Llama-3 8B reference run specified in the design remain to be executed. The conclusions are correspondingly scoped: the dissociation is established for this construction on these models, and its generality is the first task of the work that follows.

The distance manipulation is narrow. Prenominal modification adds at most two tokens, a constraint imposed by the need to avoid the attachment ambiguities that postnominal modifiers would introduce, and a distance-sensitive process could in principle be flat over two tokens and visible over ten. The human digging-in studies used substantially longer regions. The equivalence analysis bounds how large an effect the present range can exclude, but extending the range without contaminating the structure, for instance with stacked adjectives or structurally inert adverbials, is a design problem worth solving, and the refutation of H1b should be read as scoped to the tested range.

The confirmatory result is a null interaction, and null results carry an asymmetric burden. The design addresses this in part: the interaction was fixed in advance as the confirmatory test, the main effect against which it is compared is large, no split half shows significant growth and the only significant split-half interaction runs toward shrinkage, the null replicates across two model families, and the equivalence analysis converts the null into a bounded claim. A larger item set would nonetheless tighten the bound, and the planned mechanistic evidence is designed to converge on the same conclusion by a route that does not run through a null.

The plausibility screen is circular in one respect: added modifiers were screened for plausibility drift using GPT-2-large, which subsequently served as one of the two analysis models. The screen could in principle have selected items whose surprisal behavior is unrepresentative for that model. Two considerations limit the damage: the screen operated on the modifiers' effect on continuation plausibility, not on

the critical-word contrast the analysis tests, and Pythia-1.4B, which played no role in screening, shows the same pattern. A rerun of the screen with a model outside the analysis set is nonetheless the clean fix and is planned before any expanded item set is frozen. Human norming of the modifiers was infeasible within the timeline and remains the stronger standard.

Training-data contamination is a standing concern for garden-path materials, since classical items recur in the literature the models were trained on. The materials here are adapted lexicalizations from the SAP Benchmark rather than classical sentences, and the distance manipulation introduces novel surface strings, but contamination cannot be excluded outright; a from-scratch model trained on a fully inspectable corpus is the clean anchor and is left to future work.

On models: the two analysis models differ in scale as well as in architecture family, training data, and tokenizer, so their agreement is evidence of robustness across model families, not evidence about scale, and no scaling claim is made here. The planned developmental work is assigned to the Pythia suite, whose released sizes share training data and data order and therefore support controlled comparisons of scale (Biderman et al., 2023). Instruction tuning shifts next-token distributions, so all present and planned analyses anchor on base checkpoints. Finally, no new human data is collected; comparisons to human difficulty run through published norms, which constrains model-human magnitude claims, and the claims of this paper accordingly concern the structure of the model effect rather than its fidelity to human magnitudes.

9 Significance and Outlook

The methodological contribution is an item design that orthogonalizes hierarchical relationship from linear distance, together with the derivation showing that the design tests the linear account twice: the presence of a garden-path effect refutes strictly local prediction, and its failure to grow refutes recency-weighted prediction over the tested range. Any account of garden-path processing in language models must now say what it predicts about distance, because the two factors no longer travel together. The empirical contribution is the dissociation itself: on two open-weights models, the garden-path effect at the disambiguating word is large and does not grow with linear distance, which removes the linear explanations of reanalysis behavior and establishes the behavioral pattern the encoded-but-not-used account requires. The theoretical contribution is the reframing this enables. The models' lack of a distance penalty diverges from human digging in, which suggests that the underprediction of van Schijndel & Linzen (2021), Arehalli et al. (2022), and Huang et al. (2024) is not a uniform shortfall: part of it appears to be a length-dependent component the models lack. The useful question then concerns timing, when structure gets consulted and what dynamics surround the misparse.

The follow-on program is future work and gets one para-

graph. Structural probes with control tasks, paired with causal activation patching, can separate the structure a modern reference model encodes from the structure it deploys, and can localize the reanalysis computation the present paper observes only behaviorally. The Pythia checkpoint ladder can chart when, over training, structure becomes decodable against when it begins to drive the output; the encoded-but-not-used account predicts a lag, narrowing with training and scale. And the human comparison sketched in Section 7, regressing the human-model gap on ambiguous-region length in the SAP reading-time data, would convert the divergence documented here into a quantitative decomposition of the underprediction puzzle. The behavioral dissociation reported here is the foundation those analyses stand on.

Acknowledgments

I used Claude (Anthropic) to help write the plotting code for Figure 1 and to check the equivalence-bound calculations reported in Section 6.2. All code was written, run, and verified by the author, and all analytical decisions and claims are my own.

References

- Ahuja, K., Balachandran, V., Panwar, M., He, T., Smith, N. A., Goyal, N., & Tsvetkov, Y. (2025). Learning syntax without planting trees: Understanding hierarchical generalization in transformers. *Transactions of the Association for Computational Linguistics*, 13, 121–141.
- Arehalli, S., Dillon, B., & Linzen, T. (2022). Syntactic surprisal from neural models predicts, but underestimates, human processing difficulty. In *Proceedings of CoNLL*.
- Biderman, S., et al. (2023). Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of ICML*.
- Ferreira, F., & Henderson, J. M. (1991). Recovery from misanalyses of garden-path sentences. *Journal of Memory and Language*, 30(6), 725–745.
- Frazier, L., & Rayner, K. (1982). Making and correcting errors during sentence comprehension. *Cognitive Psychology*, 14(2), 178–210.
- Futrell, R., Wilcox, E., Morita, T., Qian, P., Ballesteros, M., & Levy, R. (2019). Neural language models as psycholinguistic subjects: Representations of syntactic state. In *Proceedings of NAACL*.
- Hale, J. (2001). A probabilistic Earley parser as a psycholinguistic model. In *Proceedings of NAACL*.
- Hewitt, J., & Liang, P. (2019). Designing and interpreting probes with control tasks. In *Proceedings of EMNLP*.
- Hewitt, J., & Manning, C. D. (2019). A structural probe for finding syntax in word representations. In *Proceedings of NAACL*.
- Huang, K.-J., Arehalli, S., Kugemoto, M., Muxica, C., Prasad, G., Dillon, B., & Linzen, T. (2024). Large-scale benchmark yields no evidence that language model surprisal explains syntactic disambiguation difficulty. *Journal of Memory and Language*, 137.
- Lakens, D. (2017). Equivalence tests: A practical primer for *t* tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4), 355–362.
- Levy, R. (2008). Expectation-based syntactic comprehension. *Cognition*, 106(3), 1126–1177.
- Linzen, T., Dupoux, E., & Goldberg, Y. (2016). Assessing the ability of LSTMs to learn syntax-sensitive dependencies. *Transactions of the Association for Computational Linguistics*, 4, 521–535.
- McCoy, R. T., Frank, R., & Linzen, T. (2020). Does syntax need to grow on trees? Sources of hierarchical inductive bias in sequence-to-sequence networks. *Transactions of the Association for Computational Linguistics*, 8, 125–140.
- Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems*.
- Michaelov, J., & Arnett, C. (2025). Disaggregation reveals hidden training dynamics: The case of agreement attraction. In *CogInterp: Interpreting Cognition in Deep Learning Models Workshop at NeurIPS 2025*.
- Misra, K. (2022). minicons: Enabling flexible behavioral and representational analyses of transformer language models. *arXiv preprint arXiv:2203.13112*.
- Murty, S., Sharma, P., Andreas, J., & Manning, C. D. (2023). Grokking of hierarchical structure in vanilla transformers. In *Proceedings of ACL*.
- Petty, J., & Frank, R. (2021). Transformers generalize linearly. *arXiv preprint arXiv:2109.12036*.
- Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language models are unsupervised multitask learners. OpenAI Technical Report.
- Tabor, W., & Hutchins, S. (2004). Evidence for self-organized sentence processing: Digging-in effects. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 30(2), 431–450.
- van Schijndel, M., & Linzen, T. (2021). Single-stage prediction models do not explain the magnitude of syntactic disambiguation difficulty. *Cognitive Science*, 45(6).